

SECOND SEMESTER DIPLOMA EXAMINATION IN ENGINEERING
AND TECHNOLOGY

PRODUCTION ENGINEERING I
MODEL QUESTION PAPER – SET-1

Time: 3 hours

Maximum Marks: 75

PART A

I. Answer all the following questions in one word or sentence.

(9 x 1 = 9 Marks)

1	What is a continuous chip?	M1.01	R
2	What is oblique cutting?	M1.02	U
3	Name any two cutting tool materials.	M1.02	R
4	What is 18-4-1 HSS?	M2.01	R
5	What is relief angle?	M2.02	U
6	What is fusion welding?	M3.02	R
7	What is a carburizing flame?	M3.02	R
8	What is flux in welding?	M3.02	U
9	What is a wing core?	M4.03	U

PART B

II. Answer any Eight questions from the following

(8 x 3= 24 Marks)

1	Explain types of tool wear.	M1.03	R
2	What are the factors affecting machinability.	M1.04	R
3	What are cemented carbides .Explain its composition	M2.01	R
4	Explain properties of cutting fluids	M2.03	U
5	What are the properties of lubricants.	M2.03	R
6	What is soldering.	M3.02	U
7	Distinguish between Gas and Arc welding.	M3.02	U

8	Explain defects in welding.	M3.03	R
9	What are core binders.	M4.03	R
10	What are the advantages of powder metallurgy	M4.04	R

PART C

III. Answer all questions from the following

(6 x 7 = 42 Marks)

1	Explain taylor's Tool Life equation	M1.03	U
OR			
2	Define cutting speed, feed, depth of cut.	M 1.02	U
3	Distinguish between side cutting edge angle and end cutting edge angle	M 2.02	U
OR			
4	Describe various cutting tool materials	M 2.01	R
5	Distinguish between hot and cold working.	M 3.03	U
OR			
6	Distinguish between smith forging and drop forging	M 3.04	R
7	Explain percussion welding	M 3.02	R
OR			
8	Explain MIG welding.	M 3.02	R
9	Explain compacting in powder metallurgy.	M 4.04	U
OR			
10	Explain various types of patterns.	M 4.01	R
11	Explain the mechanical properties of parts made by powder metallurgy.	M 4.04	U
OR			
12	Describe the types of cores used in pattern making.	M 4.03	U

SECOND SEMESTER DIPLOMA EXAMINATION IN ENGINEERING
AND TECHNOLOGY

PRODUCTION ENGINEERING I
MODEL QUESTION PAPER – SET-2

Time: 3 hours

Maximum Marks: 75

PART A

I. Answer all the following questions in one word or sentence.

(9 x 1 = 9 Marks)

1	What is chip thickness ratio?	M1.01	R
2	What is orthogonal cutting?	M1.02	U
3	What are the 3 types of chips produced in metal cutting?	M1.02	R
4	What is 18-4-1 HSS?	M2.01	R
5	What is rake angle?	M2.02	U
6	What is plastic welding?	M3.02	R
7	What is an oxidizing flame?	M3.02	R
8	What is arc length?	M3.02	U
9	What is a wing core?	M4.03	U

PART B

II. Answer any Eight questions from the following

(8 x 3 = 24 Marks)

1	Explain Taylor's Tool Life equation.	M1.03	R
2	Explain machinability index	M1.04	R
3	Explain properties of cutting fluids	M2.03	U
4	Explain characteristics of ideal tool materials.	M2.01	R
5	What are the factors affecting the choice of cutting fluids.	M2.03	R
6	What are the functions of coating in electrodes	M3.02	U
7	Distinguish between spot and seam welding.	M3.02	U

8	Explain defects in welding.	M3.03	R
9	Explain types of cores in pattern making.	M4.03	R
10	Explain the properties of moulding sand.	M4.02	R

PART C

III. Answer all questions from the following

(6 x 7 = 42 Marks)

1	Explain types of tool wear.	M1.03	U
OR			
2	Distinguish between orthogonal and oblique cutting.	M 1.02	U
3	Distinguish between rake angle and relief angle	M 2.02	U
OR			
4	Describe various cutting tool materials	M 2.01	R
5	Distinguish between hot and cold working.	M 3.03	U
OR			
6	What are the types of forging.	M 3.04	R
7	Explain thermit welding	M 3.02	R
OR			
8	Explain TIG welding.	M 3.02	R
9	Explain sintering in powder metallurgy.	M 4.04	U
OR			
10	Explain various types of patterns.	M 4.01	R
11	Explain the advantages and disadvantages of powder metallurgy.	M 4.04	U
OR			
12	Describe the types of cores used in pattern making.	M 4.03	U